Optical properties of gold-silica-gold multilayer nanoshells

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Abstract: The spectral and angular radiation properties of gold-silica-gold multilayer nanoshells are investigated using Mie theory for concentric multilayer spheres. The spectral tunability of multilayer nanoshells is explained and characterized by a plasmon hybridization model and a universal scaling principle. A thinner intermediate silica layer, scaled by particle size, red shifts the plasmon resonance. This shift is relatively insensitive to the overall particle size and follows the universal scaling principle with respect to the resonant wavelength of a conventional silica-gold core-shell nanoshell. The extra tunability provided by the inner core further shifts the extinction peak to longer wavelengths, which is difficult to achieve on conventional sub-100 nm nanoshells due to limitations in synthesizing ultrathin gold coatings. We found multilayer nanoshells to be more absorbing with a larger gold core, a thinner silica layer, and a thinner outer gold shell. Both scattering intensity and angular radiation pattern were found to differ from conventional nanoshells due to spectral modulation from the inner core. Multilayer nanoshells may provide more backscattering at wavelengths where silica-gold core-shell nanoshells predominantly forward scatter.

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1. Introduction

The human use of gold has been well documented throughout history. Observations have been long made on the unique optical properties of nano-sized particles composed of gold. The rapid development in controlled chemical synthesis and computational modeling has generated an extensive exploration on various gold-related nanostructures, ranging from nanorods [1] and nanostars [2, 3], to core-shell nanoshells [4–6], nanorice [7], and nanocages [8]. Silica-gold core-shell nanoshells have attracted particularly significant research attention due to agile optical tunabilities [4, 9] and absorption and scattering cross sections which can greatly exceed those of organic dyes [10, 11]. Superior photochemical stability in combination with excellent biocompatibility renders gold nanoshells highly attractive for biomedical imaging and spectroscopy applications, such as optical coherence tomography [12], reflectance spectroscopy [13], dark-field and two photon microscopy [11, 14, 15]. Gold nanoshells have also been investigated for photothermal therapy and controlled drug release [16, 17].

Extinction spectra of the core-shell nanoshell can be tuned by varying the gold-shell thickness scaled by particle size. The plasmon resonance of gold nanoshells is related to the interaction between plasmons supported on the inner and outer surface of the gold shell. The strength of their interaction is determined by the shell thickness scaled by particle size. As the gold shell decreases in thickness, a stronger plasmon interaction red shifts the resonance peak compared to that of a solid gold particle. This allows the tuning of nanoshell plasmon resonances into the near-infrared (NIR) region [18] where main biological absorption is minimal [19]. The universal dependence of red shifts upon shell thicknesses scaled by particle size was reported by Jain *et al* [20].

Here we examine the optical properties of gold-silica-gold multilayer nanoshells. Fig. 1. illustrates the structure of multilayer nanoshells and silica-gold core-shell nanoshells. Xia *et al*. were the first to report the synthesis of $\sim$50 nm multilayer nanoshells that may exhibit NIR absorption peaks [21]. Coating gold colloid with a thin layer of silica was achieved by a modified Stöber method in which silica growth was preceded by a sodium silicate (active silica) treatment in an aqueous solution with a controlled pH. The outer gold coating was produced similarly to the way conventional nanoshells are made. Due to the use of small gold colloids ($\sim$20 nm) and the relative ease of silica coating with various thicknesses, sub-100 nm multilayer nanoshells can be synthesized. While comparable in size with some solid gold spheres, a greater spectral tunability is expected for multilayer nanoshells due to the interaction of plasmons on interfaces between gold and dielectrics. Such nanoshells could also offer a smaller profile than their conventional counterparts, thus provide better vascular permeability and more efficient antibody conjugation owing to the larger surface-to-volume ratio [21].

Thus far, the optical properties of the multilayer nanoshell, composed of a metallic core and two alternating dielectric and metallic layers, have been investigated by Chen *et al*. to achieve ultrasharp resonant peaks across the spectrum for multiplexing applications [22]. In the study, the overall diameter of the nanoshell was kept at 10 nm and the layers were tailored with

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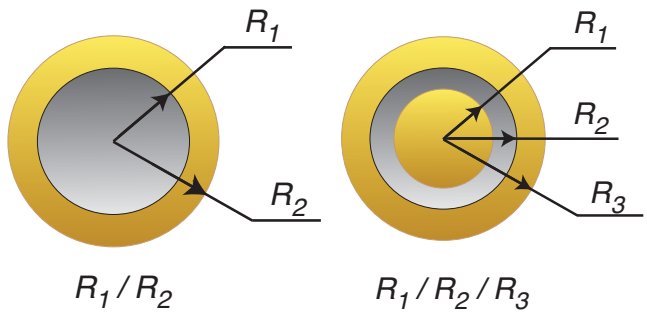
subnanometer precision. The thin layers sparked some controversy over spectral broadening due to the intrinsic size effect on metal properties for nano-sized particle simulations [23].



(a) Conventional nanoshell (CNS) (b) Multilayer nanoshell (MNS)

Fig. 1. Geometries of (a) silica-gold core-shell conventional nanoshells, and (b) gold-silica-gold multilayer nanoshells.

The goal of this paper is to examine the spectral and angular scattering properties of gold-silica-gold multilayer nanoshells in the size region where successful particle syntheses have been reported and can be achieved based on currently available protocols [21, 24]. The study is based on what has been done on conventional nanoshells, with the objectives of unveiling and understanding the similar and distinct optical properties of multilayer nanoshells, as well as investigating their potential for bioimaging applications.

2. Methodology

A Mie-based computation code has been developed to calculate light scattering from concentric spheres [25–27]. Boundary conditions at each inter-layer interface are expressed in term of vector spherical wave functions and unknown field coefficients. The tangential electric and magnetic fields are stored as alternating rows for each interface in a square matrix and the vector of unknown coefficients is resolved by a matrix division. This approach has a substantial computational advantage as it can handle subjects from spheres with stratification to spheres with gradient index profiles. Extinction, scattering and absorption coefficients were calculated based on Mie scattering coefficients obtained from the Mie code. Angular radiation was calculated for unpolarized light using the approach outlined by van de Hulst [27]. Details of the code can be found in [25].

In this study, plane-wave expansion coefficients are used for illumination. This assumes that particles are located far enough from the light source or that the particles are significantly smaller than the incident beam profile. Other assumptions include that the particles are rigidly spherical and the layers are concentric. It is to be noted that various factors, such as surface topology [28], core eccentricity [29] and interparticle distance [30] have been reported to affect the nanoshell spectra to various degrees.

For gold properties, data from Cristy and Johnson [31] are adapted for the simulation. While the spectral broadening effect from surface scattering in nano-sized particles has been proposed and observed on gold colloids [32,33], the current literature on intrinsic effects of the core-shell nanoshells has not been able to reach a consensus. Common practice at times adds an additional term to the bulk material dielectric constant to acount for the limited free electron path imposed by the thin gold shell [4]. Others propose that this modification is determined by the core radius to shell thickness ratio [34]. However, it was reported that intrinsic effects were not observed

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on spectral measurement of single silica-gold nanoshells [35]. In light of controversies over this issue, intrinsic size corrections were not considered in this study. In addition, we attempt to construct gold layer geometries outside the region where intrinsic effects prevail.

In simulations presented, water is the surrounding medium unless otherwise noted. The dielectric constant for silica was set to 2.04 and that for water was 1.77. The Mie code was validated against published spectral results for gold particles [36], silica-gold nanoshells [36], and gold-silica-gold-silica nanoshells [37], as well as angular radiation patterns of silica-gold nanoshells [38].

3. Spectral properties

3.1. Tunability from the inner gold core

In this section, effects from the inner gold core are qualitatively explained by a plasmon hybridization model and quantitatively examined using Mie theory. The goal is to explore the possibilities of synthesizing multilayer nanoshells with enhanced optical properties in the NIR region.

Similar to conventional nanoshells (CNS), multilayer nanoshells (MNS) have tunable optical properties, as explained by plasmon hybridization theory [37]. Briefly stated, the tunability of a CNS is attributed to the interaction between plasmons that reside on the outer and inner surface of the gold shell, also known as the sphere and cavity plasmon. The interaction causes the plasmon to split into a low-energy bonding mode and a high-energy anti-bonding mode. The bonding mode is often visualized by the surface plasmon resonance peak of the nanoshell in the vis-NIR region. The peak can be tuned by varying the ratio of shell thickness to core radius, which essentially tunes the coupling strength between the two plasmons.

In contrast to CNS, MNS have an extra degree of tunability from the inner gold core. This optical tunability can be understood as an interaction between the CNS bonding mode $|BN\rangle$ and the gold core sphere mode $|rC\rangle$. The thickness of the intermediate silica layer determines the degree of interplay between the two modes. An increase in the inner gold core radius on an otherwise fixed geometry will decrease the intermediate silica layer thickness and increase the plasmon interaction. This is accompanied by a red shift of the spectrum that is in agreement with Mie calculation results, shown in Fig. 2 (Media 1).

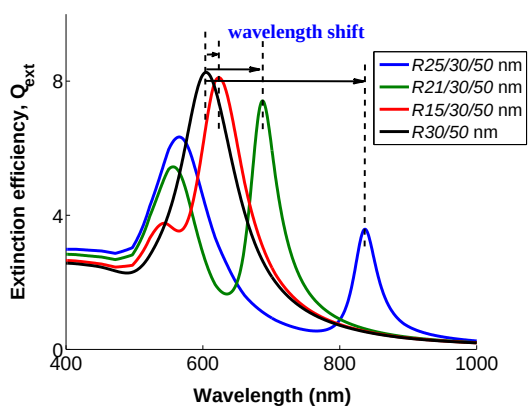

Fig. 2. Calculated spectra of CNS and MNS with various inner core radii while the silica and outer radius remain the same (Media 1). The red shift of MNS from CNS is indicated by lambda shift.

The extra tunability introduced by the inner gold core facilitates the synthesis of small MNS

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with NIR extinction peaks. This is not an option for CNS. Near-infrared extinction is difficult to achieve on small CNS because extremely thin outer gold layers are required. The current coating process involves functionalizing the silica core surface with amine groups, then attaching small gold colloids (1–2 nm in size) to form nucleation sites for further reduction of gold to form a continuous layer [6]. However, for sub-10 nm coats, it is often difficult to achieve even layers and smooth surfaces, both of which are needed to avoid drastic deterioration in the overall integrity of the nanoshell spectrum [28, 29]. For MNS, these requirements are alleviated because extinction peaks can be further red shifted by having a larger inner gold core with a thin surrounding silica layer. The compromise is a damped peak owing to the increasing plasmon strength associated with the gold core mode $|rC\rangle$.

3.2. *Universal scaling principle*

Given that the plasmon coupling strength is qualitatively determined by the intermediate silica layer thickness, a quantitative analysis is desired to characterize the spectrum signature of MNS with various dimensions. Among recently reported observations is the universal scaling principle, which was first reported for metal particle pairs in which plasmon resonance is red shifted from that of an isolated metal sphere by moving the two particles closer together [39]. The shift exhibits a near-exponential decay with increasing interparticle distance. Jain et al. extended this theory to CNS where the gold shell thickness, scaled by particle size, is analogous to the interparticle gap and the two interacting plasmon modes act as a particle pair [20]. A similar exponential decay is observed independent of the overall nanoshell dimension. While MNS support more than two plasmons, their plasmon resonance is mainly determined by the interaction between the bonding mode of CNS and the sphere mode of the gold core. The CNS anti-bonding mode, however, has a very small dipole moment because the cavity plasmon is oppositely aligned with the sphere mode. For this reason, the CNS anti-bonding mode interaction with the core mode is small and becomes too damped to be visible in the spectra. The coupling strength between the two major-playing modes is determined by the intermediate silica layer thickness, scaled by the overall particle size. With R_2/R_3 ratio kept constant, a universal decaying curve is observed for MNS of different dimensions.

Figure 3 illustrates an exponential decay insensitive to the overall particle size at a constant ratio of $R_2/R_3 = 0.6$. Results (data not shown) indicate that both the R_2/R_3 ratio and the surrounding medium affect the rate of decay. In addition, larger MNS were found to retain the same exponential decay but at slightly different rates. This can be partially attributed to the fact that multiple resonant peaks start to emerge with broad widths and less well-defined shapes. Nevertheless, the universal scaling principle demonstrates that CNS plasmon resonant peaks can be further red shifted on MNS by reducing the silica layer thickness. For particles retaining their overall dimension and outer shell thickness, this translates to the use of large gold cores with thin silica coatings.

It is worth noting that when t is pushed to the zero limit where $R_1 = R_2$, thus the silica layer thickness approaches zero, the plasmon mode red shifts to zero energy. However, the spectral weight becomes negligible due to the cancellation of the dipole moments of the core and of the shell.

3.3. *Sensitivity to the surrounding medium*

The plasmon hybridization model can be used to explain the spectral sensitivity of MNS to the surrounding medium. MNS plasmons are resulted from the hybridization between CNS bonding modes $|BN\rangle$ and gold core sphere modes $|rC\rangle$, and since $|rC\rangle$ is not in direct contact with the outside, $|BN\rangle$ is the mode mainly affected by the surrounding medium. For instance, an increase in the refrative index of the medium does not directly affect $|rC\rangle$ but red shift $|BN\rangle$,

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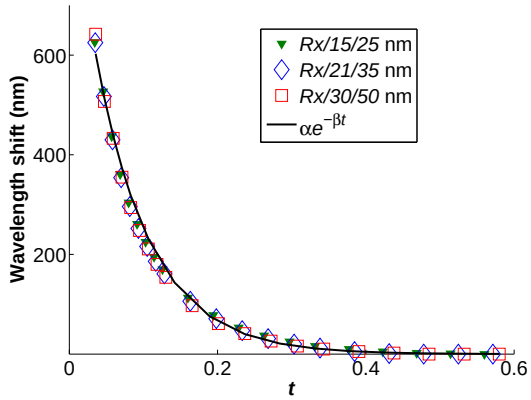

Fig. 3. Mie calculation results for wavelength shifts from the MNS ($R_1/R_2/R_3$) vis-NIR plasmon resonance peak relative to that of CNS (R_2/R_3) with the same silica and outer gold radius versus intermediate silica thickness scaled by particle size of MNS. $t = (R_2 - R_1)/R_3$, α = 972.43, β = 13.48.

contributing to an overall red shift as demonstrated in Fig. 4. This suggests that MNS may be used as sensors in various sensing applications based on detecting localized surface plasmon resonances (LSPRs).

While CNS and other nanostructure have been studied for sensing applications [40,41], MNS may offer some unique opportunities given multiple extinction peaks shifted in a synchronous fashion when the surrounding medium is perturbed. Moreover, the degree of shift of each peak was found to be particle-dependent. Table 1 tabulates the wavelength shift for two MNS with an identical size and gold shell thickness. Between the two, MNS2 bears a stronger plasmonic coupling between $|BN\rangle$ and $|rC\rangle$ than MNS1. It can be seen that MNS2 has a larger peak shift at the shorter wavelength and its inter-peak distance decreases with increasing medium indices. The opposite was found for MNS1. Also observed is that the shift of each peak of the MNS approximately adds up to that of the single SPR on an equivalent CNS.

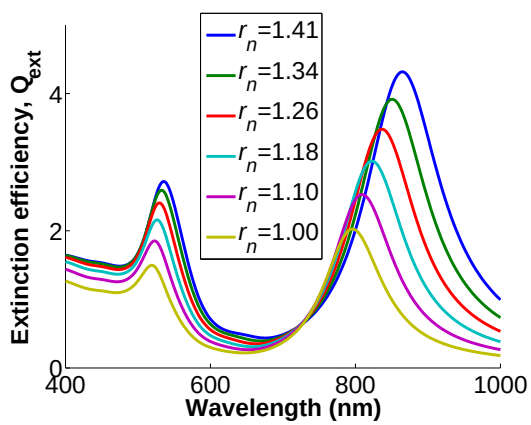

Fig. 4. Calculated extinction spectra of $R20/30/35$ nm MNS immersed in various media with distinct refractive indices.

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Table 1. Surface plasmon resonance peak shift of $R20/35/50$ nm (MNS1), $R30/35/50$ nm (MNS2), and $R35/50$ nm (CNS) in different dielectric media ($\lambda_{p2} > \lambda_{p1}$).

	p1 shift (nm)		p2 shift (nm)		p2-p1 (nm)		p shift (nm)
refractive index	MNS1	MNS2	MNS1	MNS2	MNS1	MNS2	CNS
1.0	0	0	0	0	110	424	0
1.1	5	8	10	4	115	420	15
1.2	11	20	22	9	121	413	33
1.3	16	37	38	15	132	402	56
1.4	21	53	56	22	145	393	80
1.5	27	68	77	29	160	385	105

3.4. Tunability on scattering and absorption

Rayleigh criteria state that, for particles much smaller than the wavelength, the scattering intensity is proportional to R^6, where R is the particle radius [42]. However, absorption depends on particle volume, which is proportional to R^3 for a sphere. The higher order dependence on particle size makes scattering more sensitive to size variations than absorption. For instance, at 532 nm, 100 nm solid gold spheres have approximately the same absorption and scattering cross sections in water [42]. Scattering attenuates more drastically than absorption when particle size decreases, rendering gold colloids absorption-dominant. In contrast, increasing particle size enhances scattering much faster than absorption; thus, large particles are mainly scattering-based. For inhomogeneous spheres, such as silica-gold core-shell CNS, it has been demonstrated that the scattering-to-absorption ratio rises with increasing core size or outer shell thickness or both [12]. These observations provide very useful guidelines for synthesizing particles to match either scattering- or absorption-based applications at desired wavelengths [16,43].

The scattering and absorption behavior of concentric multilayer spheres can be studied similarly as functions of layer thickness to obtain insight into how changes in each layer affect the overall scattering and absorption. Due to the complexity in displaying results with concurrent variation in each of the three layers, simulations were adapted to geometries in which two layer thicknesses were changed while the third layer remained fixed. Three sets of runs were performed with three combinations of the two variable-layer thicknesses. At each dimension, the extinction, scattering, and absorption spectra were calculated from 400 to 1941 nm. Two plasmon resonance wavelengths λ_{max} were chosen at the extinction maxima. The scattering to extinction ratio was calculated at the longer of the two; the one associated with the MNS mode rather than the gold sphere mode. If only one maximum occurred in the wavelength range, then the calculation was performed at that wavelength. Since the resulting plots illustrate the scattering ratio at the longer plasmon resonant wavelengths, we expect the magnitude of the extinction peak to attenuate as it red shifts and it may become weaker than the resonance at shorter wavelengths.

3.4.1. Inner gold core radius fixed at 10 nm

This inner core radius was chosen based on the first experimental synthesis of MNS [21]. Figure 5(a) indicates that scattering increases with thicker silica layers or thicker outer gold shells or both. This behavior resembles that reported for CNS [12]. With a thin silica layer coating (<10 nm) [21, 24, 44], MNS exhibit more absorption than scattering. However, when the silica layer thickness is increased beyond 20 nm, MNS become mainly scattering at practical outer shell

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thicknesses.

3.4.2. Intermediate silica layer fixed at 10 nm

For these calculations, Fig. 5(b) shows that the general trend still holds that thicker outer shells produce more scattering; however, the role of the inner gold core is less clear. Below 20 nm, an increase in the core radius slowly increases the scattering-to-extinction ratio, but this trend seems to reverse when the core radius goes above 20 nm. Another set of simulations on 20 nm silica layers (not shown here) indicate minimal effects from the gold core below 10 nm in radius and the scattering-to-extinction ratio became relatively insensitive to further increases in core radius. At thin silica layer thicknesses ($<$5 nm), MNS behavior is similar to solid gold colloids in that increasing the inner core radius gradually enhances scattering.

3.4.3. Outer shell fixed at 20 nm

Similar to the results in Fig. 5(a) and 5(b), Fig. 5(c) displays a profile in which absorption is dominant or equivalent to scattering for thin silica layers ($<$10 nm). Thicker silica layers ($>$10 nm) quickly turn MNS into scattering particles, with minimal impact from the inner gold core.

3.4.4. Outer shell and overall size fixed

To elucidate the core effect, the overall diameter and the outer shell thickness were kept fixed. The inner core radius was gradually increased to the point at which the core made contact with the outer shell. The absorption-to-extinction ratio was obtained at the plasmon resonance, as described above, and results are shown in Fig. 5(d).

Four different geometries with the same R_1/R_2 ratio are plotted in Fig. 5(d). As the core radius increases, the absorption component at the plasmon resonance also increases. This is counterintuitive since we expect large gold colloids to exhibit more scattering than absorption. The explanation lies in the interplay between the gold core and other layers of the MNS. For instance, an increasing core size results in a thinner silica coating in this otherwise fixed geometry, and according to Fig. 5(a)–5(c), thinner silica layers produce less scattering; this is consistent with the behavoir in Fig. 5(d). Furthermore, the discontinuities at large R_1/R_2 ratios reflect the region where the silica layer becomes so thin that the MNS resonant wavelengths lie outside the region of interest and the gold colloid resonance dominates. MNS demonstrate strong scattering characteristics associated with solid gold spheres. It can be seen that large particles (i.e. $Rx/80/100$ nm) produce more scattering than small particles (i.e. $Rx/24/30$ nm). Because one of the main advantages of MNS over CNS is their relatively small size owing to the use of small gold cores and the possibility of coating them with thin silica layers, sub-100 nm MNS are perceived as advantageous in absorption applications where NIR extinction is desired.

4. Angular radiation properties

To understand scattering enhancement by nanoshells in applications where detection angle and angular acceptance range may differ, angular properties need to be considered. Angular properties are commonly characterized by the overall radiation power and its directivity within certain angular ranges. The former is defined by the scattering cross section, C_{sca}, and the latter is often described by a single-value parameter: the anisotropy factor g, which is a cosine-weighted average over all values of the scattering angle. The scattering cross section is calculated from the Mie coefficients [27]. The anisotropy factor g is calculated as

$$g = \frac{\int_0^\pi S_{11}(\theta) cos\theta 2\pi sin\theta d\theta}{\int_0^\pi S_{11}(\theta) 2\pi sin\theta d\theta} \tag{1}$$

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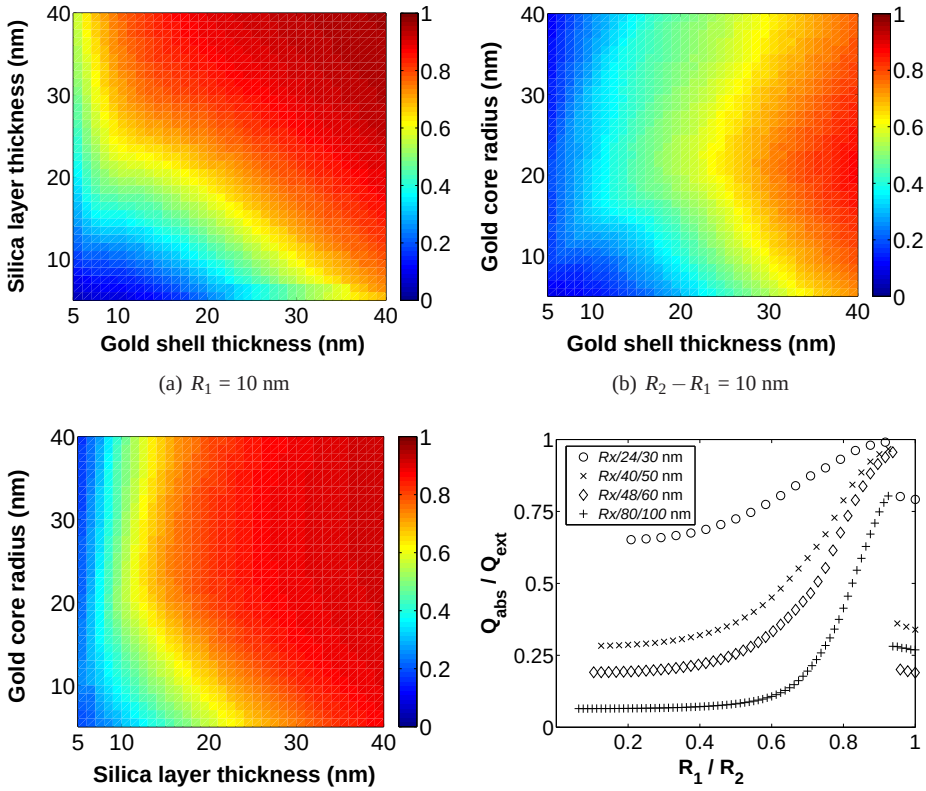

(a) R_1 = 10 nm

(b) $R_2 - R_1$ = 10 nm

(c) $R_3 - R_2$ = 20 nm

(d) R_3 fixed

Fig. 5. Scattering-to-extinction ratio at plasmonic resonant wavelengths of MNS with (a) inner gold core radius kept at 10 nm and varying the silica and outer gold layer thickness, (b) silica layer thickness kept at 10 nm and varying the inner gold core radius and outer gold layer thickness, (c) outer gold shell kept at 20 nm and varying the inner gold core radius and silica layer thickness. Absorption-to-extinction ratio (d) at plasmonic resonant wavelengths of MNS with the overall diameter and outer gold shell thickness fixed and varying the inner gold core radius and silica layer thickness.

where S_{11} is the angular radiation power of unpolarized light.

For strongly scattering nanoshells ($>$ 200 nm in size), the main extinction peak coincides with the scattering peak where the radiation pattern grows to an overall maximal level. Because MNS and CNS exhibit distinct spectra, as shown in Fig. 2, drastically different radiation patterns are expected at some wavelengths. Figure 6 (Media 2) compares the radiation patterns from MNS and CNS computed as the radiation power normalized to the incident power at one meter from the center of the particle. The radiation power was integrated over all azimuthal angles and plotted in logarithmic form: $\log_{10}(P_{rad}/P_{inc})$. Scattering cross sections are highlighted at selected wavelengths corresponding to the radiation pattern. The overall size and outer shell thickness were chosen so that the corresponding CNS are scattering-dominant.

It can be seen that $R90/125/140$ nm MNS scatters more at 550 nm [Fig. 6(a)], whereas $R125/140$ nm CNS radiates more at 755 nm [Fig. 6(b)] and 1145 nm [Fig. 6(c)]. At 1270 nm the two nanoshells scatter approximately the same [Fig. 6(d)]; this is also indicated by the nearly equivalent scattering cross sections [Fig. 6(e)]. Although optical cross sections give an

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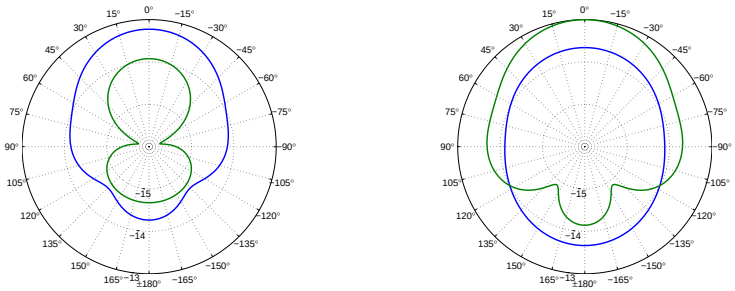
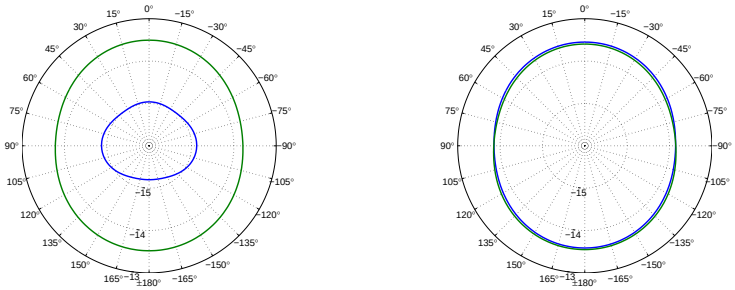


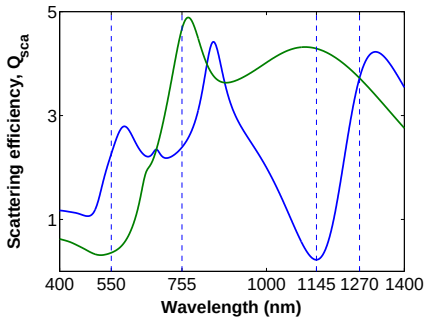
(a) λ =550 nm

(b) λ =755 nm



(c) λ =1145 nm

(d) λ =1270 nm



(e) Q_{sca}

Fig. 6. Angular radiation pattern of R90/125/140 nm MNS (blue) and R125/140 nm CNS (green) at (a) 550 nm, (b) 755 nm, (c) 1145 nm, (d) 1270 nm and (e) scattering spectra of MNS (blue) and CNS (green) (Media 2). Nanoshells are located in the center of the plot, and the incident wave enters from the bottom (180°).

overall indication of the radiating power, they do not provide angular properties. For instance, despite an overall stronger radiation at 755 nm, $R125/140$ nm CNS does not project as much power in the back direction as $R90/125/140$ nm MNS. From the radiation patterns above, it can also be observed that $R125/140$ nm CNS is more forward-scattering (positive g) at shorter wavelengths and becomes isotropic ($g \sim 0$) and slightly back-scattering (negative g) at wavelengths longer than the resonance wavelengths. The $R90/125/140$ nm MNS, however, does not exhibit regularities that can be associated with spectral signatures.

To further consolidate this observation, spectral properties are plotted side by side with the anisotropy factor at different wavelengths for 200 nm CNS and MNS with various layer geome-

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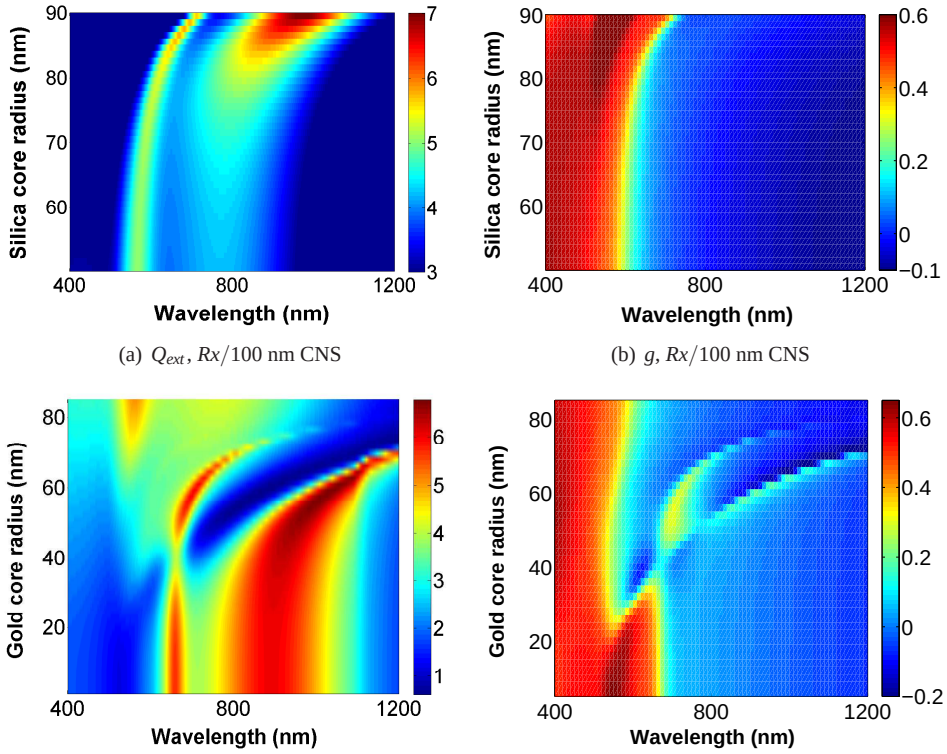
tries in Fig. 7. A well-defined border between large and near-zero g values is found to follow the general trend of CNS spectra. This suggests the plasmon resonant wavelength is a boundary beyond which CNS primarily scatter isotropically. Nevertheless, MNS seem to reach low g values at shorter wavelengths prior to the plasmon resonance. This may indicate a stronger back scattering profile compared to CNS, as already seen in Fig. 6(b).



(a) Q_{ext}, $Rx/100$ nm CNS

(b) g, $Rx/100$ nm CNS

(c) Q_{ext}, $Rx/85/100$ nm MNS

(d) g, $Rx/85/100$ nm MNS

Fig. 7. (a) Extinction efficiency spectra for $Rx/100$ nm CNS with varying silica core radius x. (b) Anisotropy factor plot for corresponding CNS at different wavelengths. (c) Extinction efficiency spectra for $Rx/85/100$ nm MNS with varying gold core radius x. (d) Anisotropy factor plot for corresponding MNS at different wavelengths.

5. Summary

Both spectral and angular radiation properties of gold-silica-gold multilayer nanoshells have been studied using Mie theory. The plasmon hybridization model was employed to explain the spectral tunability due to inner gold core. While the plasmon coupling strength of CNS is known to be determined by a normalized gold shell thickness, that of the MNS was found to be determined by the normalized intermediate silica layer thickness. A thinner silica layer results in a red shift of the plasmon resonance. Furthermore, the MNS spectral shift from CNS without the gold core is characterized by an exponential curve that was found to be insensitive to the particle size, when the outer shell thickness-to-particle size is fixed. This confirms the universal scaling principle reported on particle pair systems and silica-gold core-shell CNS. MNS also demonstrate many characteristics that are similar to CNS. For instance, a thicker

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silica layer and a thinner gold shell both red shift the MNS spectrum (results not shown) and produce more scattering at the plasmon resonance. MNS are sensitive to the external medium and their multiple extinction peaks with deep spectral valleys may prove valuable for improving the sensitivity and specificity of various biosensing and bioimaging modalities.

Scattering patterns from MNS differ from those of CNS due to the spectral modulation induced by the core. Trends in MNS angular radiation patterns are more intricate than those of CNS. While CNS predominantly forward-scatter at wavelengths shorter than the plasmon resonant wavelength, MNS may radiate more in the back and side directions at these wavelengths.

In summary, this study compares the optical properties of composite multilayer structures with those of CNS, whose properties are well understood. Fully exploiting the potential of MNS requires synthetic studies assessing independent control of each layer, morphological and topological features, size dispersity, and protocol repeatability. It is anticipated that such studies will lead to development of multilayer nanoshells, which, in turn, will accelerate the development of new applications.

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